The Paleomagnetism of the Permian Red Beds of Southwest Tanganyika

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Abstract. Samples have been collected from twelve sites in red horizons of lower Permian (Ecca) age in southwest Tanganyika. Two directions of magnetization are observed, one steeply inclined with north-seeking poles down and the other with south-seeking poles down in the direction of the earth's present magnetic field. Specimens from all the sites were subjected to both alternating-field and thermal demagnetization. These laboratory studies show that the component which is north-seeking down with steep inclination is the stable direction of magnetization. It is suggested that the component aligned along the earth's present field is due to the growth of hematite (CRM) in this field. After treatment at temperatures of 300°C, specimens from nine of the twelve sites gave well-grouped directions of magnetization. On the assumption of a geocentric axial dipole field, the mean pole position is at 38.2°S, 64.9°E (A_{95} = 16°). This pole position places southern and eastern Africa in temperate or cold-temperate climatic zones, which is in accord with the available paleoclimatic evidence. The polarity (reversed) of the Ecca of southwest Tanganyika is in accord with all other Permian rocks that have previously been studied.

Introduction. During the 1962 field season specimens were collected from red horizons of Ecca age (lower Permian) from the Galula coalfield, the Songwe-Kiwira coalfield, and the Ketewaka-Mchuchuma coalfield. Farther to the south in Africa the rocks of Ecca age do not contain red horizons, and attempts to obtain paleomagnetic information for this region have been unsuccessful [*Nairn,* 1960; *Graham,* 1961].

Field work was severely limited by the lack of roads, which limited accessibility to outcrops. Artificial exposures such as road cuts and quarries are practically nonexistent. The most useful exposures proved to be the beds of dry rivers, where erosion during the wet season served to expose fresh rock. Some specimens were actually cored under water because the rocks tend to disintegrate rapidly when exposed to weathering.

Geology. The Karroo sediments of southwest Tanganyika are preserved in basins downfaulted into the Precambrian basement rocks. The biggest of these is the Ruhuhu basin (3 in Figure 1), which contains the type section of the Tanganyika Karroo. In the Ruhuhu basin the Karroo rocks have been divided into eight rock units ranging in time from Dwyka (upper Pennsylvanian) to Beaufort (upper Triassic) [*Stockley,* 1932; *Stockley and Oates,* 1931]. These eight rock units are designated K1 through K8. This paper deals with the K3 known in the type area (Ketewaka-Mchuchuma coalfield of the Ruhuhu basin) as the Intermediate marls and sandstones.

In the Ketewaka-Mchuchuma coalfield the K3 consists predominantly of green or buff sandstones and grits; at the top of the succession red mudstones are intercalated between the sandstones. These mudstones present formidable sampling problems, since they are predominantly unindurated and disintegrate rapidly on exposure. The area has suffered little tectonism except normal faulting since the deposition of the beds, and the structural dips are low [*McKinlay,* 1954].

The K3 succession of the Songwe-Kiwira (2 in Figure 1) coalfield consists of massive feldspathic sandstones near the base of the section with red, and red and green mudstones and siltstones in the upper part of the formation. The coalfield has been subjected to faulting, probably in connection with the formation of the Nyasa rift, and the beds dip at angles

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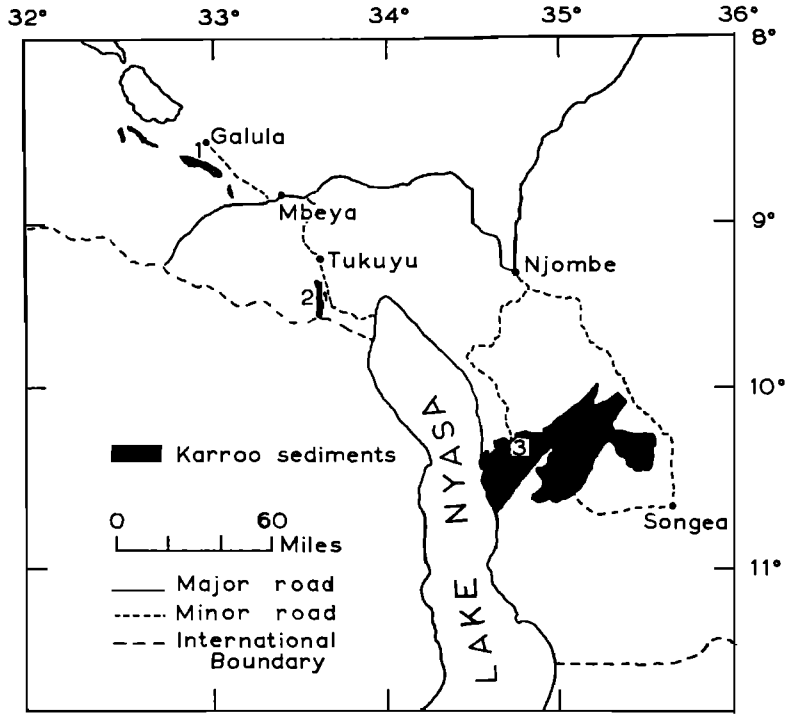


Fig. 1. Sketch map of southwest Tanganyika showing some Karroo coalfields. 1, Galula coalfield; 2, Songwe-Kiwira coalfield; 3, Ketewaka-Mchuchuma coalfield.

between 20° and 30° to the east [*Harkin*, 1955].

The K3 horizon of the Galula coalfield (1 in Figure 1) consists of an alternating series of massive feldspathic sandstones with interbedded red siltstones. No mudstones were observed in the section. The coalfield lies along the edge of the Rukwa rift valley and has been subjected to normal faulting which causes the beds to dip at angles up to 30° to the north [*Spence*, 1954].

The K3 in each of these fields lies conformably upon the K2 coal beds which are Lower Ecca in age. The age of the K2 coals has been determined from an abundant *Glossopteris* flora. In the type area, coals also containing a *Glossopteris* flora overlie the K3 beds. It seems unlikely that the beds studied can be younger than middle Permian; however, the precision of this correlation rests upon the accuracy with which the Karroo formations are correlated with their northern hemisphere equivalents.

Hart [1960], on the basis of a microflora assemblage extracted from the Lower coal measures (K2) of the Ketewaka-Mchuchuma coalfield, has correlated these beds with the Permian of Australia. The beds in the Australian succession with which the K2 beds are considered to be wholly or in part equivalents are the Greta coal measures of New South Wales, the Holmewood shale of the Irwin basin, the Poole sandstone and uppermost Grant formation of the Fitzroy basin of Western Australia. He considers the K2 of the Tanganyika succession to be not younger than middle Artinskian.

Field sampling. More than a hundred oriented cores were collected from six sites in the Galula coalfield, from two sites in the Songwe-Kiwira coalfield, and from four sites in the Ketewaka-Mchuchuma coalfield. The cores were drilled with a *Graham and Keiller* [1960] drill, and an azimuthal orientation accuracy of 1° was obtained by the method of *Gough and Opdyke* [1963]. Each site can be considered a point in time, since the stratigraphic thickness sampled at any site was of the order of 3 m or less. The sites in the Galula coalfield are in the gorge of the Elephant River and cover a stratigraphic interval of approximately 250 m. In the Songwe-Kiwira coalfield the two sites

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are in the bed of the Illima River in the Illima outlier near an access road to the Illima Colliery. The two sites are separated by approximately 100 m stratigraphically. The sites in the Ketewaka-Mchuchuma coalfield were sampled from the beds of the Mchuchuma and Nyakalai rivers. The stratigraphic separation of these sites is unknown, but they are from separate horizons. Since the sites were sampled in three different areas several hundred miles apart it is felt that the sampling has been adequate in space and time. *Nairn* [1964] has previously sampled the section from the Songwe-Kiwira coalfield.

During this field trip I also visited the Nagaka coalfield of the Ruhuhu basin and the Mhukuru coalfield. In the Nagaka coalfield suitable exposures could not be found. Many of the sections could not be visited, however, and more extensive field work might meet with success. In the Mhukuru coalfield the red mudstones of the K3 were located, but the exposures were so badly weathered that no samples were obtainable.

Natural remanent magnetization. The directions of natural remanent magnetization for two sites are shown in Figures 2 and 3. The data have not been corrected to the horizontal. In Figure 2 magnetic directions are present in the same sense as the earth's present field (normal) and in the opposite sense (reversed), in almost exact opposition to each other. In Figure 3 the directions are scattered, and the direction for some specimens is near that of the present field. At sites 2, 5, 7, and 9 the directions form well-defined groups which in their present attitude are almost 180° reversed from the earth's present field.

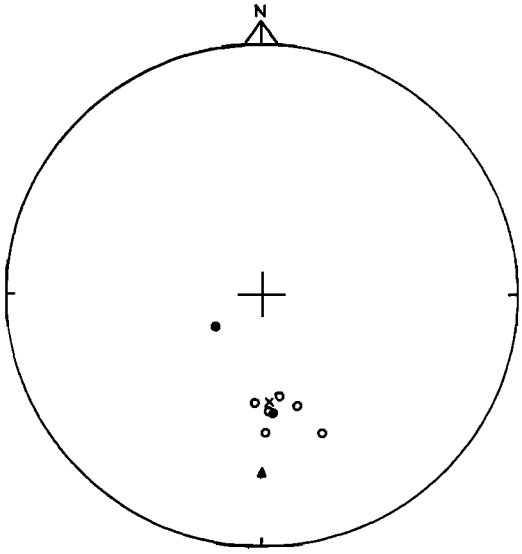


Fig. 2. NRM directions of magnetization from a site in the Galula coalfield. Lower-hemisphere stereographic projection. Open circles, south-seeking direction down; closed circles, north-seeking direction down; triangle, present axial dipole field direction (south down); cross, direction of the earth's present magnetic field (south down).

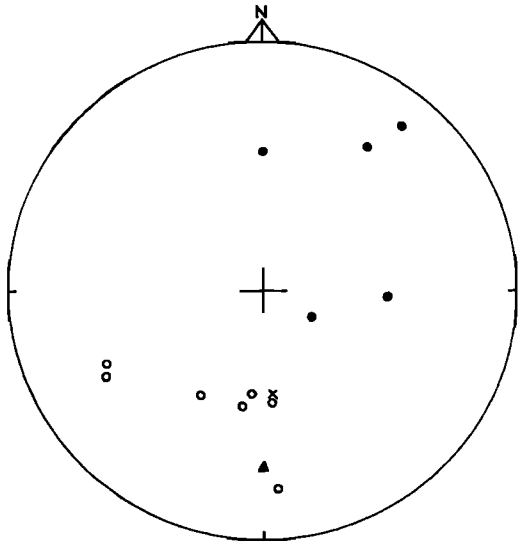


Fig. 3. NRM directions of magnetization from site 8. Lower-hemisphere stereographic projection. Symbols are as in Figure 2.

Alternating-field demagnetization. Alternating-field demagnetization [*Thellier and Rimbert,* 1954; *As and Zijderveld,* 1958; *Creer,* 1959] was attempted on the samples from all the sites; equipment and techniques described by *McElhinny and Gough* [1963] were used.

The effect of alternating-field demagnetization at site 8 is examined below in some detail. Figure 3 shows the distribution of NRM for 12 cores taken from a stratigraphic interval of not more than 70 cm from the bed of the Mchuchuma River. The change of direction on treatment in alternating fields of several cores can be seen in Figures 4 and 5. Specimen *A* (Figure 4) moves only about 20° up to 300 gauss and then stays in roughly the same position in peak fields of up to 800 gauss. Specimen *B* (Figure

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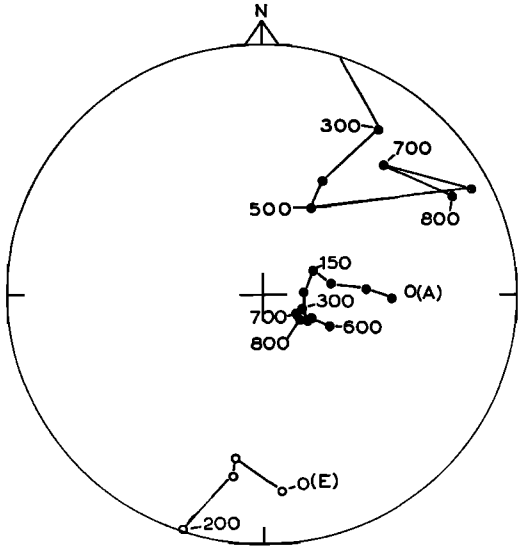


Fig. 4. Change in direction of specimens *A* and *E*, site 8, after successive demagnetization in the peak alternating fields indicated (gauss). Lower-hemisphere stereographic projection. Open circles, south-seeking direction down; closed circles, north-seeking direction down.

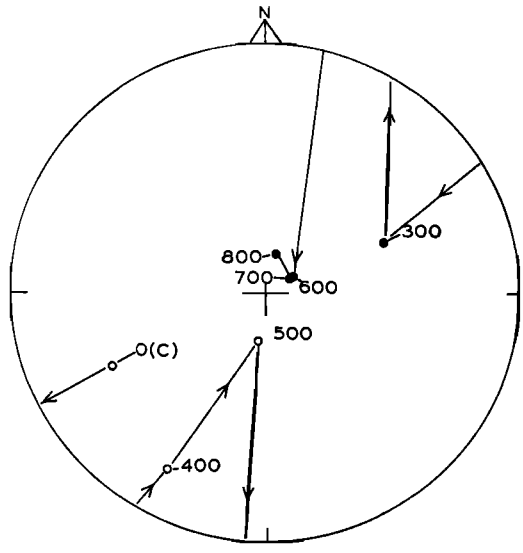


Fig. 5. Change in direction of specimen *C*, site 8, after successive demagnetization in the peak alternating field indicated (gauss). Lower-hemisphere stereographic projection. Open circles, south-seeking direction down; closed circles, north-seeking direction down.

6) moves hardly at all on treatment, so the change in direction is not shown. When the magnetic moment is plotted against increasing field (Figure 6) the intensity of these two specimens never drops below 75 per cent of the NRM, and on application of higher fields it actually becomes stronger than the original moment. Points *C* and *E*, which have polarities of the same sign as the present earth's field, become reversed in polarity. Specimen *E* (Figure 4) approaches the stable direction given by specimen *A* but never reaches a stable end point. Specimen *C* (Figure 5) shows rather startling changes in directions. The initial direction is normal; it becomes reversed at 300 gauss and reverts to normal again in fields of 400 and 500 gauss. It then reverses through almost 180° and remains stable in a steeply inclined reversed direction in fields of 600, 700, and 800 gauss. This shows that the stable component is reversed in sign to that of the present field and that the directions grouped around the direction of the earth's present field are spurious. The normalized demagnetization curves for specimens *C*, *D*, and *E* are markedly different from those of *A* and *B* (Figure 6), which changed direction very little. For these specimens the intensities fall off rapidly to about 20 per cent of the NRM in fields up to 300 gauss and then change very little in increasing fields.

Figure 7 shows demagnetization curves for other sites from stable and partially stable rocks. In some partially stable rocks the moment increases steadily until it is 80 per cent or more above its NRM value. Others increase or decrease slightly but remain within 20 per cent of the original moment.

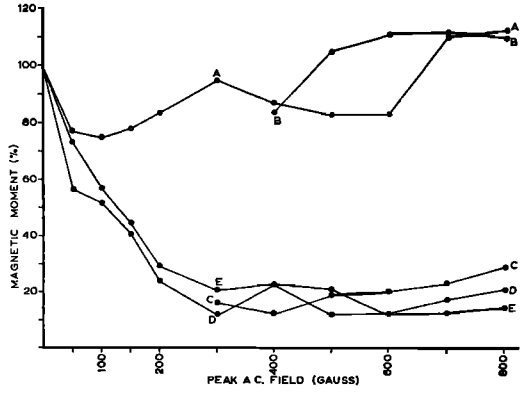


Fig. 6. Alternating-field demagnetization curves of specimens from site 8.

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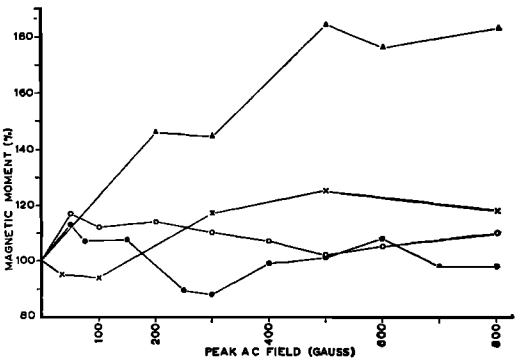


Fig. 7. Alternating-field demagnetization curves from stable or partially stable specimens.

Thermal demagnetization. An oven for heating specimens to temperatures up to 800°C has been constructed in this department. The oven, being smaller than the one described by *Irving et al.* [1961], will hold only six 2.5-cm specimens at a time. A bank of four furnaces will soon be in operation, giving both volume and flexibility. The smaller size of the oven allows the furnace to heat to 700°C in about an hour, which greatly increases the speed of operation. The furnace is positioned in a set of three orthogonal pairs of square Parry coils [*Parry*, 1957] approximately 180 cm on a side. The earth's field at the furnace can be canceled while the specimens cool. The operating procedure is very similar to that described by *Irving et al.* [1961] except that specimens are heated in air.

Thermal demagnetization curves on six specimens from six different sites are shown in Figure 8*a*. All of these specimens had an NRM in the same direction as the earth's present field. The moment of all specimens, with the exception of no. 2, fell off very sharply in low temperatures. Specimens 1, 3, 5, and 6 changed in direction (in same cases by almost 180°) to steeply inclined north-seeking poles down at temperatures ranging from as low as 200°C for specimen 5 to 550° for specimens 3 and 1 (Figure 9). At 630°C specimen 2 had retained its polarity (south-seeking down), whereas specimen 4 became north-seeking down at 550°C but never became steeply inclined. The temperature needed to remove the unstable component is variable both within specimens from a site and between specimens from different sites. Specimen 5 exhibited curious behavior at high temperatures. It reversed at 200°C and

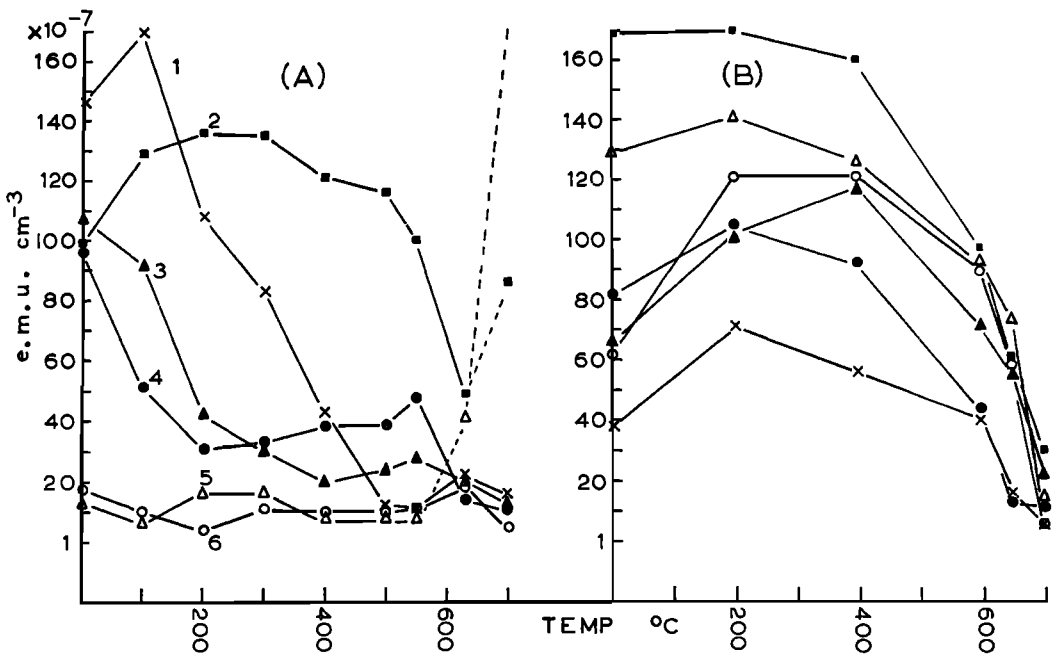


Fig. 8. Thermal demagnetization curves of (*A*) partially stable and (*B*) stable specimens.

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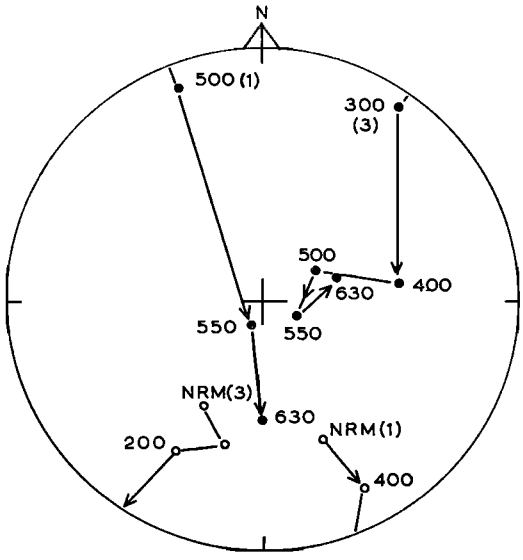


Fig. 9. Change in direction of specimens 1 and 3 after successive thermal demagnetization in temperatures (°C) as indicated. Lower-hemisphere stereographic projection. Open circles, south-seeking direction down; closed circles, north-seeking direction down.

remained stable up to 550°C. At 630°C the moment dramatically increased, and on remeasurement 3 hours later, the direction had changed and second harmonic was visible on the monitoring oscilloscope [*Gough*, 1964]. The specimen was placed in the earth's field for 3 days and on remeasurement was found to have accurately assumed the direction of the earth's field. On heating and cooling above the Curie point of hematite the specimen again increased its moment dramatically. This behavior was probably due to a chemical change of an unknown nature. It is unlikely to have been due to the formation of magnetite since it was heated in air. Specimen 1 also increased its moment between 650° and 700°C. In general the specimens of Figure 8*a* show 'thermally distributed components' [*Irving and Opdyke*, 1964], and the blocking temperatures are as low as 100°C.

Specimens which were initially steeply magnetized with the north-seeking pole down were thermally demagnetized again, and the demagnetization curves are shown in Figure 8*b*. These specimens exhibit stable remanent magnetization but do not have a distinguishable thermally discrete component, [*Irving and Opdyke*, 1964].

Samples from all sites were treated at temperatures of 300°C or more. In most cases the directions of magnetization became closely grouped with steeply dipping north-seeking poles down (sites 9, 2, 5, 3, 4, 7). Other sites (8, 6, 1), when treated in 300°C, became partially grouped, although some outstanding specimens still retained an appreciable component in the direction of the earth's present field. In many cases this was removed by heating to higher temperatures. This behavior is expected because the specimens previously illustrated in Figure 9 'cleaned' at different temperatures. Three sites were discarded because a majority of specimens from these sites maintained their direction of magnetization in the direction of the earth's present field. At each of these sites individual specimens changed their polarity on heating to north-seeking poles down.

Comparison of alternating-field demagnetization with thermal demagnetization. After the extensive heating and alternating-field demagnetization experiments it becomes possible to attempt an explanation of the observed behavior of these specimens.

1. NRM directions at some of the sites exhibit 'streaking' between a reversed direction and the earth's present field.

2. On treatment in alternating fields and on heating and cooling in zero field, partially stable specimens and some that could be considered to be unstable moved toward or joined the reversed group.

3. The reversed group with steep inclinations exhibited stable behavior in alternating fields and on thermal demagnetization.

4. At no time did a reversed specimen change to the polarity of the earth's present field.

The reversed direction of magnetization is a stable direction of magnetization, and the direction is fairly uniform from site to site; therefore, it is reasonable to presume that this is the original Permian magnetization. The specimens which give intermediate directions of magnetization, probably representing an intermediate stage in the growth of hematite grains, have small grain diameters, and the grains contribute only a small component in the direction of the earth's field. This is probably a form of viscous magnetization.

These observations are in accord with the

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ideas of *Creer* [1961] which were based on the theory of TRM for single-domain grains put forward by *Néel* [1955]: The grain size of hematite is very critical in determining whether the directions will be stable. Very small grains are unstable, but the stability increases as the grain size increases until a critical blocking diameter is reached above which the hematite possesses a very stable magnetization. As these specimens are subjected to severe conditions of heavy rainfall and drought, hematite will begin to grow in the earth's present field, and cleaning techniques might be effective when the grain size is small. When the grain size grows larger it seems improbable that alternating-field treatment will be effective, because of the high coercivity of hematite. Similarly, thermal treatment is ineffective as the blocking temperature approaches the Curie point. The end product of this process would be the complete remagnetization of the beds in the direction of the present magnetic field.

In the treatment of partially unstable red beds where the remanent magnetization is due to CRM, thermal treatment is by far the more powerful technique of the two used here in removing the later CRM. Specimens which are unaffected by alternating fields of up to 800 gauss will reverse their direction upon thermal treatment in temperatures from 300° to 550°C. Even specimens which show well-grouped NRM improve their grouping on thermal treatment. In this study thermal treatment allowed the recovery of sites which were, on first sight, utterly useless. Figure 10 illustrates the directions at one such site before and after thermal treatment; this site failed to respond to alternating-field treatment. I strongly recommend that paleomagneticians studying the magnetization of red beds employ thermal techniques. These have been successfully used in this study and in previous studies of Silurian red beds in North America [*Irving and Opdyke*, 1964], Devonian red beds in England [*Chamelaun and Creer*, 1963], and Australian sediments [*Irving et al.*, 1961; *Irving*, 1963]. In Table 1 a comparison of the directions before and after thermal treatment shows the remarkable increase in k for many of the sites.

Pole position. The mean pole position for the Permian of southwest Tanganyika is calculated from the site poles in Table 1. Assuming a geocentric axial dipole field, the mean north pole position is 38.2°S, 64.9°E, $A_{95} = 16°$. Figure 11 shows that the site directions fall into two groups, one with a marked easterly component representing those sites from the Songwe-Kiwira coalfield, and the Ketewaka Mchuchuma coalfield. The sites from Galulu coalfield have directions more to the south (Figure 11). These

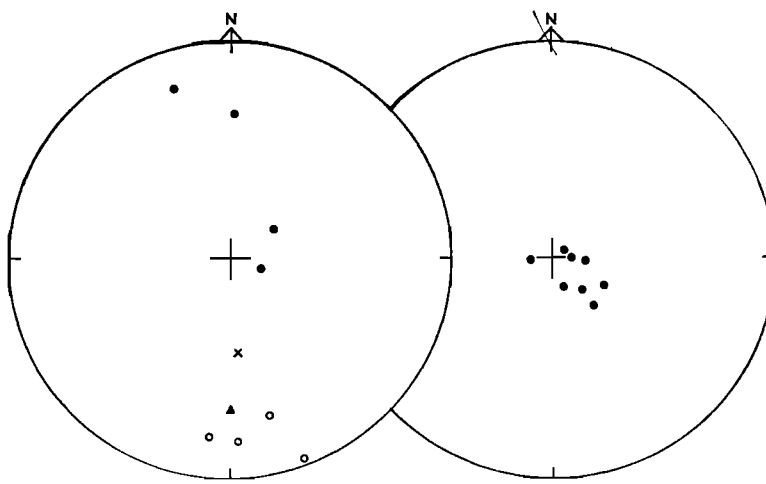


Fig. 10. Directions of magnetization at site 6; (left) NRM and (right) after thermal demagnetization at 300°C. Lower-hemisphere stereographic projection. Open circles, south-seeking direction down; closed circles, north-seeking direction down; triangle, present axial dipole field direction (south down); cross, direction of the earth's present magnetic field (south down).

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TABLE 1. Site Directions for the Permian of Southwest Tanganyika

Locality	Site No.	N	E	M_0 D	M_0 I	M_0 R	M_0 k	M_{th} °C	M_{th} D	M_{th} I	M_{th} R	M_{th} k	M_{th} α_{95}	Site Position Lat. S	Site Position Long. E	Site Pole Lat. S	Site Pole Long. E
Galula coalfield	1	8	1			2.82*		300	156½	72½	7.910	36.8	9.2	8°50′	32°56′	38°	48½°
	2	5	0	165½	62½	4.96	110	300	165½	64½	4.990	404	3.8	8°50′	32°56′	50½°	48½°
	3	7	1			2.85*		300	177½	73	6 928	83.4	6.6	8°50′	32°56′	40½°	34½°
	4	5	0			1 58*		300	185	66	4.983	326	5.0	8°50′	32°56′	50°	28°
	5	9	0	190½	71½	8 82	43.9	400	172	68	8.936	125	4.6	8°50′	32°56′	40½°	47°
Songwe-Kiwira coalfield	6	8	2			6.10*		300	95	58	7.757	28 8	10 5	9°30′	33°40′	10°	86°
	7	4	0	114	55	3.99	308	300	125½	46	3.990	312	5 2	9°30′	33°40′	36°	96½°
Ketewaka-Mchuchuma coalfield	8	7	6			5.06*		300	112½	57	6 966	176	4 6	10°20′	34°40′	18°	85°
	9	8	0	118	55	7.86	48.9	300	130	55	7.991	777	2 0	10°20′	34°40′	38°	87°

M_0 gives the NRM results; M_{th} gives those after thermal demagnetization in the temperature indicated.
N is the number of samples (i.e. cores) included in the analysis and E is the number excluded.
R is the length of the resultant of N unit vectors and D and I are the declination and inclination of its direction.
α_{95} is the semiangle of the cone of 95 per cent confidence, and $k = (N - 1)/(N - R)$ is the best estimate of Fisher's dispersion parameter.
* The resultant R is less than the value R_0 tabulated by *Watson* [1956] which is exceeded in only 5 per cent of the cases by the resultant of N randomly directed vectors.

two groups are apparent before the directions of magnetization have been corrected for the attitude of the beds. The increase in k on restoring the beds to the horizontal is not significant. This deviation can be due to sampling inadequacies or to a genuine movement of the pole and errors in correlation between the coalfields. It is most probable that the magnetization of all sites occurred in the Permian, and most likely in the lower Permian; however an upper Permian age for the Galula sites cannot be ruled out at the present time. The polarity of the sites is in accord with all other Permian results available in the literature with the single exceptions of the Upper Tartarian of Russia [*Irving and Parry,* 1963].

The paleoclimatic implications of the Permian pole position from Africa. It has long been maintained by southern hemisphere geologists that during the Ecca stage of the Karroo system southern Africa lay in high latitudes. The reason for this contention is that the Ecca is the rock unit which is deposited on the glacial tillites, varves, and glacial pavements of the Dwyka glacial horizon. The reality and intensity of the Dwyka glaciation in Africa has seldom been questioned and is well documented [*du Toit,* 1954; *Flint,* 1961]. The Dwyka glaciation is remarkably extensive, and glacial tills and varvites have been recorded from the Congo, Southern Rhodesia, Northern Rhodesia, Tanganyika, and South-West Africa, as well as from the well-known localities of the Republic of South Africa. African geologists have been forced by this evidence to postulate that their deposition occurred at higher latitudes than

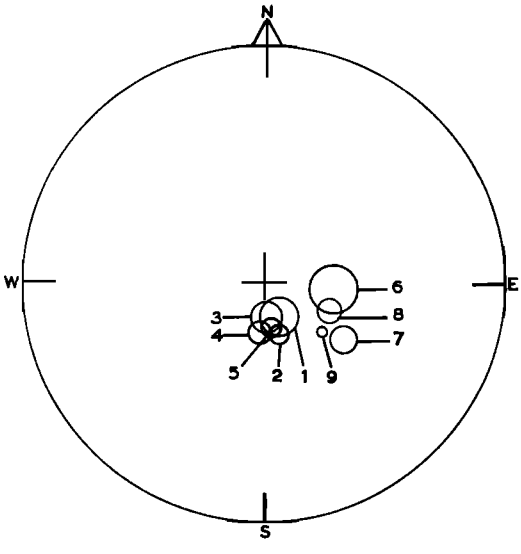


Fig. 11. Mean directions of magnetization at 8 sites, numbered as in Table 1, from the Permian of southwest Tanganyika. Circles of 95 per cent confidence are plotted on a lower-hemisphere stereographic projection.

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those in which they are found. Since many of the localities now lie well within 15° of the equator this is easy to understand. Because of its close stratigraphic association with Dwyka glaciation, the Ecca has also been assigned a position in upper or middle latitudes [*du Toit*, 1954]. If the common assumption of the coincidence of the geographic pole and the mean geomagnetic pole is accepted, this work provides strong support to this hypothesis.

The Ecca pole position (Figure 12) derived from the results presented in this paper would place those parts of Africa which were glaciated within 40° of the pole. During the Pleistocene of North America continental glaciers advanced farther than this from the geographic pole. At the same time areas of northern Africa, for instance Algeria, Morocco, and Tunisia, lay well within the tropics.

The Permian of southern Africa was the great period of coal formation; consequently the flora of this era is well known. This flora is part of the *Glossopteris* flora which is present on all of the southern continents, including Antarctica and India. Northern hemisphere elements are missing or are only a minor constituent [*Plumstead*, 1962].

There are good reasons for believing that this flora grew in a temperate climate: 1. Remains of the *Glossopteris* flora have been found interbedded with or beneath glacial horizons in both South Africa and Australia [*du Toit*, 1954]. 2. *Plumstead* [1962] presents strong evidence that the *Glossopteris* flora of South Africa was deciduous and that many of the plant-bearing beds represent autumnal accumulations of leaves.

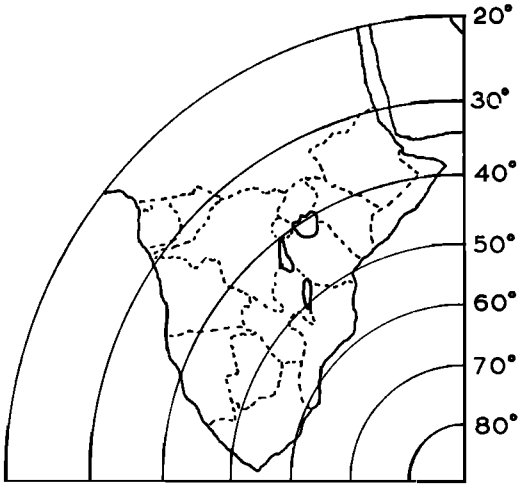


Fig. 12. The paleolatitudes of Africa during the lower Permian.

Axelrod [1963] attacked this position, as well as the validity of paleomagnetic data, on two counts:

1. He stated (p. 3260): 'However, Carboniferous swamp forests very similar to those in the northern hemisphere . . . also occupied South Africa, southeast Australia, and India, all of which were situated near or within the antarctic circle on the basis of paleomagnetic evidence.'

This statement seems difficult to reconcile with the following:

(*a*) Coal formation in southern and central Africa began in the Permian.

(*b*) Since presumably Axelrod was comparing the *Glossopteris* flora with the Pennsylvanian flora it is well to point out that most paleobotanists regard the *Glossopteris* flora to be distinct from European and American floras of the same age [e.g. *Kräusel*, 1961; *Plumstead*, 1962; *Andrews*, 1961; *Darrah*, 1960; *Edwards*, 1955; *Walton*, 1953; *Seward*, 1933; *Arnold*, 1947].

(*c*) Axelrod seems to ignore the fact that the *Glossopteris* flora has a pole-to-equator distribution in the southern hemisphere [*Plumstead*, 1962].

Hamilton [1964] in a discussion of Axelrod's paper raised this point, and in his reply *Axelrod* [1964] stated that the *Glossopteris* flora was temperate and existed on highlands alongside a tropical flora said to be the *Gigantopteris* flora. It should be pointed out that the *Gigantopteris* flora is unknown in Africa, and a mechanism has to be imagined whereby the highland temperate flora was selectively preserved while the lowland tropical flora was totally destroyed, a proposition which seems to be entirely untenable.

2. Using the pole position determined by *Nairn* [1960] from the Taru Grits of Kenya, Axelrod pointed out that this implies very large movement of Africa between the upper Carboniferous and the lower Permian. This is inconsistent with the paleobotanical evidence,

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since the *Glossopteris* flora makes its appearance in the Dwyka and is very well developed in the Ecca, persisting into the Beaufort. Nairn's pole position was of a reconnaissance nature only, and it is unlikely that his sampling represented a sufficient period of time to exclude error due to secular variation. Also, no demagnetization experiments were undertaken, and consequently some doubt must be cast on the validity of his results [*Gough et al.*, 1964]. The results presented here are in agreement with the botanical evidence, since it places southern and central Africa in high latitudes, and no large shift in latitude is necessary between the Carboniferous and lower Permian.

Climatic indicators other than floral seem to be largely lacking from the Permian of Africa. However, evaporites which are expected to develop in middle or lower latitudes are known from the Permian of the Arabian foreland. The Arabian shield is generally regarded as part of the African shield, having only recently been separated from it. These evaporites lay in tropical latitudes at the present time, and the Permian pole would likewise place them in low latitudes.

The Permian pole position deduced from this study is not in conflict with the paleoclimatic evidence but is, on the contrary, highly consistent with it.

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